Simulation and Modeling

Lecture 02

Simulation Program Development

Muhammad Mahbub Alam
Professor
CSE Department, IUT

Models

- A model is a simplified representation of some real object or physical situation which serves a particular purpose
  – Abstract representation of systems
  – A number of approximations
  – Mathematical description of the system
- Input-output model
- State space-model

State-Space Models (1/4)

System State and State Variable

- Definition (State): Complete characterization of the system at an instance of time
  - Comprehensive snapshot of the system
- Definition (State Variable): Quantitative characterization assigns numerical values to a set of variables
  - That set of variables are defined as state variables

State-Space Models (2/4)

Events

- Definition (Event): Occurrence of actions that may change the system state
  - System state cannot be changed without an event
- An event might not change the system state, but without an event system state never changes

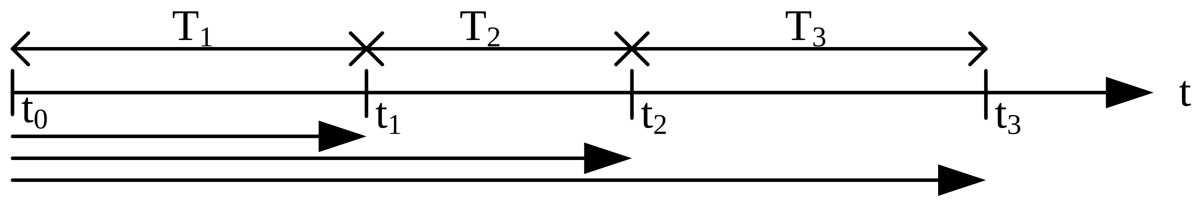
State Space

- Set of all possible values of the state variables
  - Set of all possible states of the system

State-Space Models (3/4)

Input variables

- Exact time-point (Absolute time)
  - Arrival time of a customer
    - Instance of time when a particular customer arrives
  - Departure time of a customer
    - Instance of time when the service of a particular customer is completed
- Duration of time since last event (relative time)
  - Interarrial time of customers
  - Service time of the customers



State-Space Models (4/4)

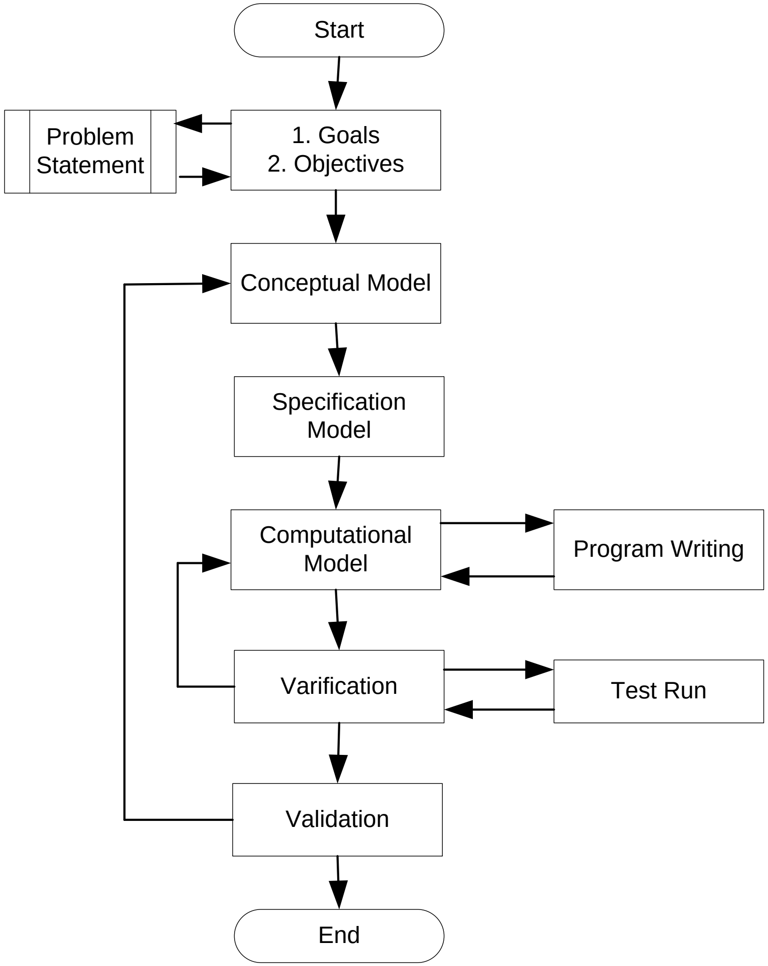
- A set of attributes represents the system model

$$\{X, E, F_x, x_0, u(t), y(t)\}$$

- X – State space of the system
- E – Set of events of the system
- F_x – Feasible set of events at state x
- x_0 – Initial system state
- $u(t)$ – Input variables
- $y(t)$ – Output variables

- Two sets of equations are derived
  - Set of state equations
  - Set of output equations

Simulation Development Life Cycle



Step 1: Goals and Objectives

- Given – Problem statement
- Goals and objectives
  - Simple Boolean decision
    - Need of an additional server
  - Numeric decision
    - # of server requires for satisfactory performance
  - Qualitative decision
    - Customer satisfaction
- Usually helps to identify the output variables

Step 2: Conceptual Model

- Usually, an informal model
- Describes the system in an abstract way
- Identifies the important variables
- Remove the irrelevant variables with negligible impact
- Intellectually challenging but rewarding

Step 2: Conceptual Model (Contd.)

- Sketch of a system, or a system diagram
- Identification of the state variables
  - How they are related
  - How they interact with each other
- Identification of the events
- Identification of the state space
- Identification of the feasible event set at a state
- Identification of the input variables
- Finding the output variables
- Relation among the variables

Step 3: Specification Model

- Mathematical representation of the system
  - Define the state equations
  - Define the output equations
    - Mathematical derivation for the output variables
    - Estimate of the output variables
- Input data modeling
  - Collection of input data
  - Statistical analysis of input data
  - Probabilistic distribution of input data
- Output Equations
  - Job-average
  - Time-average

Specification Model (1/4)

Input Data Modeling

- Event occurrence time of each of the events
  - How the time value is calculated
    - Relative Timing
    - Absolute Timing
  - How the data is provided, i.e., source of data
    - Sample data collection
    - Random data generation

Specification Model (2/4)

Input Data Modeling – Absolute Timing

- Exact timing w.r.t the simulation start time
- Need to know the timing information before starting the simulation
  - Data must be available before simulation starts
- Sort the data
- Runtime data generation is not possible
  - Uncertainty about the data size

Specification Model (3/4)

Input Data Modeling – Relative Timing

- Time gap/interval between the $(i-1)th$ and i-th occurrence of an event
  - Interarrival time of the events
  - No need to sort the data
  - Do not need all the data at the beginning
  - Runtime data generation is possible

Specification Model (4/4)

Input Data Modeling – Source of Data

- Sample data
  - Might not be available
  - Difficult to collect
  - Difficult to predict the required data size
- Randomly Generated data
  - Easy to generate
  - Probabilistic distribution is used
  - Runtime data generation is possible

Step 4: Computational Model

- Algorithm development
- Simulation clock, and a procedure to update the simulation clock
- An initialization procedure for the system
- A scheduler is designed which maintains the feasible events
  - Trigger the next event
  - Calls the simulation clock update procedure
  - Calls the event handler of the next event
- Associated with each event, a set of actions are involved – event handler executes them
  - Algorithm development for the event handler of each event

Computation Model (1/7)

Objects/Components of the Model

- Objects of the Simulator
  - Simulation clock
  - Event list
    - List maintenance mechanism
- Tasks of the Simulator
  - Time advance mechanism
  - Event Scheduling
- Objects of the system
  - State variables
  - Set of events
  - Output variables
  - Input variables

Computation Model (2/7)

Conceptual Model

- State Variables
- Set of Events
- Input Variables
- Output Variables

Specification Model

- State Equations
- Output Equations
- Input Data Modeling

Computational Model

System

- State Variables
- Events
  - Event Handling Alg
- Input variables
  - Random generator
- Output Variables
  - Output Var update Alg

Simulator

- Clock
  - Update mechanism
- EventList
- Scheduler
  - Run ()
  - Trigger Events
  - Call the handler

Computation Model (3/7)

Simulation Clock

- A variable that keeps track of the time
  - Stores the current value of the simulated time
  - Unit of time is not important
- Synchronizes the occurrence of different events
- Ensures that events occur in appropriate time
  - Event occurrence is guaranteed without breaking the order

Computation Model (4/7)

Time Advance Mechanism

- Principal approaches are
  - Next Event time advance
    - Clock value jumps from one event occurrence times to another
    - Next event among the feasible events are identified
      - Clock is advanced to the next event occurrence time
    - Time advance and event occurrence are dependent
  - Fixed increment time advance
    - Time is incremented by a fixed amount
    - At each time point event occurrence is checked
      - Event occurrence is independent of the clock updating
    - Unnecessary waiting in between the events
      - Simulation time increases

Computation Model (5/7)

Event List

- List of feasible events of current system state
- For complex systems, the list might have many entries
  - Efficient list management decreases the simulation time
- Usually, the list is kept sorted according to the event occurrence time
  - The first event will be occurred next
- Insertion of an event, sort the list again

Computation Model (6/7)

Event Scheduling

- Scheduling an event by inserting it into the event list
- The time at which the event will occur/trigger is set
  - Time period between the event scheduling and triggering is known as the event lifetime/active time
- The event handler is called
  - The function that deals with state changes associated with the event occurrence
  - Known as state transition function

Computation Model (7/7)

- Scheduling Procedure
  - Step 1: Remove the first event from the event list
  - Step 2: Update the simulation clock to the triggering event occurrence time
  - Step 3: Update the system state according to the state transition function
    - Update the state variables
    - Update all statistical variables
    - Update the trace files
  - Step 4: Update the event list
    - Add any missing feasible event
      - Usually, the triggering event is added
    - Delete any infeasible event
  - Step 5: Sort the event list
  - Step 6: if the event list is non-empty, goto step 1, otherwise, stop

Step 5: Verification

- Checking that the
  - computational Model is consistent with the specification model
  - Program is consistent with the computational model
- Algorithm design and program writing are correct
- Test run of the simulation program with known data
  - May be desk run
- Repeat steps 4 and 5, as long as not verified

Step 6: Validation

- Is the model consistent with the system
  - Model is correctly representing the system
  - Otherwise, go to step 2
- Simple but popular technique
  - Collect output data from real system for known input
  - Run the simulation with the same input
  - Collect output data from simulation
  - Consult an expert
  - If expert cannot identify the simulation and real data
    - Simulation is correct

Thank You